

1
2 Personalized AI assistants could either promote or hinder the
3 emergence of collective intelligence

4
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9 **Abstract**

10 Advances in artificial intelligence (AI) have enabled the wide use of personalized
11 assistance for cognitive tasks, which enhanced individual performance on various
12 fields. However, the impact of AI assistance on collective intelligence remains poorly
13 understood. AI assistance may undermine the emergence of collective intelligence by
14 reducing the individual diversity or introducing structured biases into a population.
15 Here, we develop a human-AI collective prediction model mediated by social learning
16 based on incentive structure by extending the existing works on collective intelligence.
17 Our model uncovers that emergence of collective intelligence is depend on how human
18 and AI interact. When AI works as if it is omniscient and gives assistance regardless
19 of users' interests, collective intelligence will be promoted. However, when AI works
20 as a chat-bot and gives assistance only based on users' interests, collective intelligence
21 could be hindered due to diversity loss and overconfidence on AI. We showed that,
22 to promote collective intelligence, penalizing or incentivizing AI usage is not reliable.
23 Instead, elaborating more diversity-aware incentives could be the key to promote and
24 maintain the collective intelligence.

25 Introduction

26 Collective intelligence (CI) describes a phenomenon in which a group’s ability to solve
27 problems, make decisions, or adapt to changing environments exceeds each member’s
28 ability [1, 2, 3, 4, 5]. This phenomenon, also referred to as wisdom of crowds, has
29 thus served as a central mechanism underlying the success of diverse organizations,
30 ranging from animal groups [6, 7, 8] and human societies [2, 9] to machine-learning
31 systems [10, 11]. Understanding the conditions that promote and maintain CI there-
32 fore remains a fundamental question across social sciences [12], evolutionary biology
33 [13], and computer science [14]. Addressing this question has become particularly
34 timely as artificial intelligence (AI) has begun to reshape how individuals acquire
35 information, make decisions, and contribute to collective outcomes [15].

36 The recent emergence of generative AIs and large language models (LLMs) has
37 changed the way individuals engage in various cognitive tasks, ranging from simple
38 everyday tasks to more complex tasks, including scientific discovery [16] and creative
39 writing [17]. In particular, while the use of generative AI models, such as ChatGPT,
40 has been rapidly increasing [18], the potential long-term impact of personal AI as-
41 sistants on human society, especially the CI, remains uncertain. Some argue that AI
42 can promote CI by enhancing individual expertise [19], and opinion aggregation [20].
43 Conversely, empirical evidence indicate that the use of AIs can limit the diversity of
44 opinions [17, 21, 22], a key requirement for the emergence of CI [23, 24, 25].

45 Our ability to predict the long-term impact of personal AI assistants on human
46 CI is further hampered by the apparent discrepancy in the incentives that shape
47 AI uses. On one hand, many industrial sectors are pushing towards increased AI
48 usages, where the number of “tokens” used is often perceived as a proxy for produc-
49 tivity [26, 27]. On the other hand, many academic institutions appear to discourage,
50 and even penalize, AI uses, reflecting concerns about plagiarism, misinformation,
51 and limited independence [28, 29]. Such discrepancy raises a question about whether
52 directly incentivizing or penalizing AI usage will benefit or harm human CI. These
53 observations further highlight the importance of identifying a robust incentive struc-
54 ture that can promote CI in the presence of personal AI assistants. Such incentive
55 structure should seek to retain the benefits of AIs and prevent potential risk, such
56 as preserving the diversity of individual opinions.

57 To address these questions, we extend previous approaches for modeling CI, where
58 each individual makes independent predictions for the outcome from partial observa-
59 tions of underlying causal factors [12, 13, 30]. In particular, we allow each individual
60 to independently consult with personal AI assistants and revise their predictions ac-
61 cordingly. The resulting model allows us to investigate the robustness of human CIs
62 under different assumptions about the AI model’s ability to make predictions as well
63 as incentive structures that are associated with the prediction task. In doing so, we
64 assess the impact of directly incentivizing or penalizing AI uses on CI and further
65 suggest a balanced incentive structure that can maintain CI in the presence of AI.

66 Main

67 Model

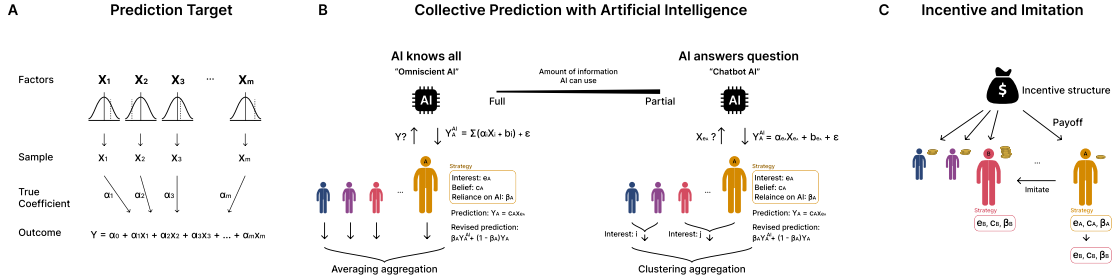


Figure 1: **A schematic diagram of the model for individual and collective prediction in the presence of personal AI assistants.** (A) The prediction target is modeled as a linear combination of causal factors. (B) Each player is characterized by three strategies: interest, belief, and reliance on AI. Players make predictions based on their interests and beliefs and subsequently revise them with AI assistance. We consider two types of AI, Chatbot AI and Omniscient AI, which differ in the amount of information available for generating predictions. (C) Players receive payoffs according to the incentive structure, and the population evolves through imitation of peers with higher payoffs.

68 We begin by extending previously established models of CI, in which a group of
 69 players participates in a prediction task (Figure 1): each player makes an individual
 70 prediction based on partial observations of the underlying causal factors, and the
 71 group forms a collective prediction by aggregating these individual predictions [30,
 72 12, 13]. Specifically, the prediction target (Y) is modeled as a linear combination of
 73 M causal factors x_i , with fixed coefficients α_i : $Y = \sum_{i=1}^M \alpha_i x_i + \alpha_0$, where each factor
 74 is drawn from independent normal distribution (Figure 1A). Then, each individual
 75 A observes a single factor of their interest x_{e_A} and makes a prediction based on
 76 their own belief c_A : $Y_A = c_A x_{e_A}$ (Figure 1B). We assume each individual can then
 77 independently consult their AI assistants and revise their prediction based on their
 78 reliance on AI β_A : $Y_A^{re} = \beta_A Y_A^{AI} + (1 - \beta_A) Y_A$ (Figure 1B). This revised prediction
 79 is aggregated to form collective prediction. **[SWP: I think we need to say averaging**
 80 **and clustering aggregation here?]**

81 To model AI, we assume that AI has already been trained on similar tasks and is
 82 able to make predictions based on partial or full observations of causal factors (Figure
 83 1B). Every time a player consults AI, it will make a new prediction Y_A^{AI} based on
 84 its internal parameters, which can include factor-specific bias (b_i) and stochastic
 85 error ϵ . Here, we consider two extreme types of AI response, which differ in the
 86 amount of information AI uses when making predictions. In one extreme, the AI
 87 can observe all causal factors and use all information to make predictions, regardless

88 of the user’s interest: $Y_A^{AI} = \sum_{i=0}^M (\alpha_i x_i + b_i) + \epsilon$. In the other extreme, AI works
 89 like a chatbot, where the AI predicts the outcome based only on the user’s interest:
 90 $Y_A^{AI} = \alpha_{e_A} x_{e_A} + b_{e_A} + \epsilon$. Hereafter, we refer to these two types as ‘Omniscient AI’
 91 and ‘Chatbot AI’, respectively.

92 Each generation consists of multiple round of predictions, and each player is re-
 93 warded a payoff according to the incentive structure (Figure 1C). For simplicity,
 94 we assume that a sufficiently large number of prediction rounds occurs within each
 95 generation, allowing us to use the expected payoff value rather than computing the
 96 payoffs for every round. We consider two incentive structures from previous work,
 97 “feedback” and “niche-expert”, which have been shown to promote near-perfect col-
 98 lective accuracy in the absence of AI [13]. The feedback structure gives higher payoff
 99 to individuals who can push the collective prediction toward the true target, regard-
 100 less of their individual predictive accuracy. [SWP: *maybe we need equations...*]
 101 In contrast, the niche-expert structure gives higher payoff to individuals who have
 102 both better predictive accuracy and a minority interest. [SWP: *maybe we need*
 103 *equations... but also say that this preserves diversity*] We note collective intelligence
 104 fails to emerge under the combination of averaging aggregation with niche-expert
 105 structure in the absence of AI [13].

106 Through time, the group evolves by allowing each individual to imitate peers
 107 who received higher payoffs (Figure 1C). Specifically, after each generation, two in-
 108 dividuals are randomly sampled, and one imitates all three strategies (the factor of
 109 interest, factor belief, and reliance on AI) of the other with a probability determined
 110 by the difference in their expected payoffs [SWP: *either refer to methods or supp.*].
 111 Individual predictions are aggregated into a collective prediction $\hat{Y}$ by simple averag-
 112 ing when using Omniscient AI and by clustering when using Chatbot AI. Collective
 113 accuracy is defined as the reduction in prediction error achieved by the collective
 114 prediction relative to a baseline prediction α_0 [SWP: *maybe need equation*]. Here-
 115 after, we refer to the near-perfect collective accuracy as accurate CI. [SWP: *How*
 116 *about: hereafter, we say that CI emerges when the collective accuracy reaches a near*
 117 *perfect level.*]

118 This model now allows us to explore the impact of AI on CI depending on the
 119 mode of AI and the incentive structure. Here, we begin by comparing the emergence
 120 of CI with and without AI and subsequently investigate a how different incentive
 121 structures affect the emergence CI in the presence of AI.

122 AI can either promote or prevent the emergence of CI

123 To understand the impact of AI, we first compare collective dynamics with and
 124 without AI. First, consistent with [13], simple imitation rules and incentive structures
 125 allow collective intelligence to emerge in the absence of AI (Fig. 2A,C,D). One
 126 exception is the use of simple averaging with niche-expert incentive structures, under
 127 which the collective accuracy of the population remains near zero [SWP: *because...*]
 128 (Fig. 2B) [SWP: *maybe need a supp figure*].

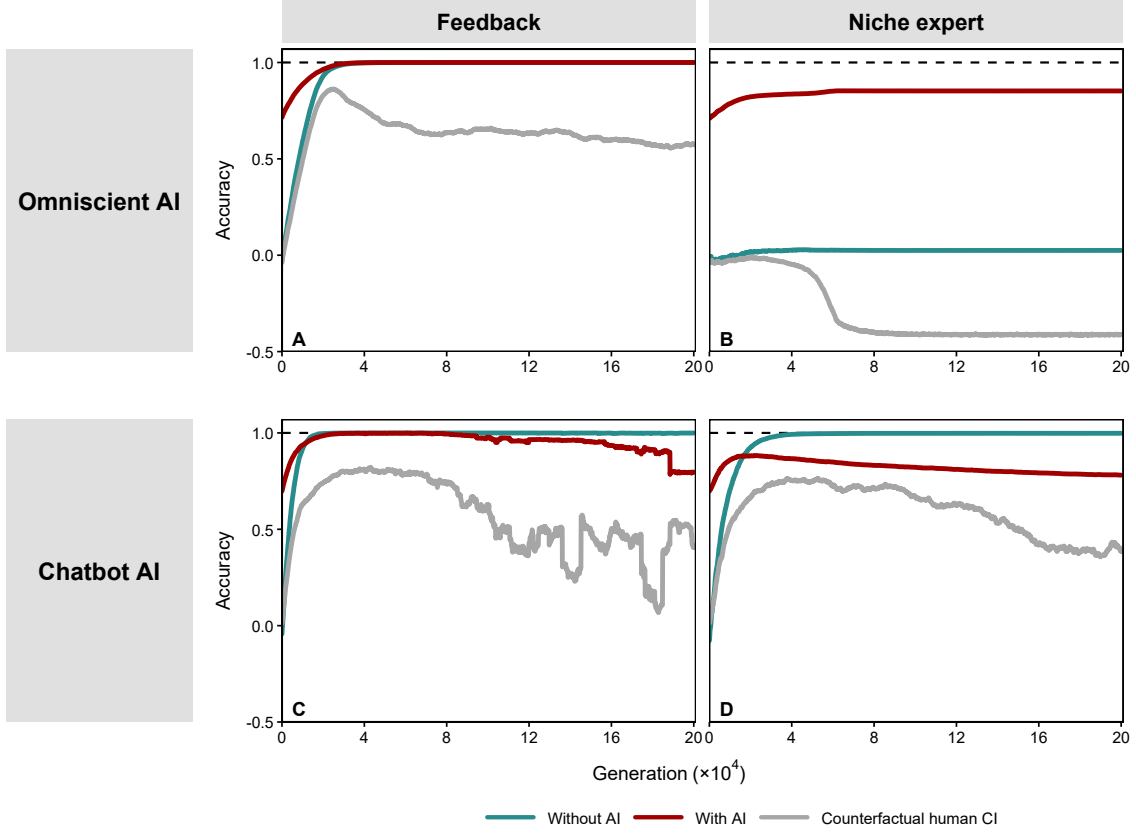


Figure 2: **Comparison of collective accuracy with and without AI.** Trajectories of collective accuracy when the group is assisted by the Omniscient AI (A,B) and Chatbot AI (C,D). [SWP: *Maybe we should also name the averaging rule below AI...*] Here, feedback (A,C) and niche-expert (B,D) incentive structures are considered. Green lines represent trajectories of collective accuracy without AI assistance. Red lines represent trajectories of collective accuracy with AI assistance. Gray lines represent trajectories of counterfactual collective accuracy, computed from players’ unrevised predictions, indicating the group’s predictive accuracy if AI assistance were unavailable at the given generation. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$ for (A, B), $c_A \sim \mathcal{N}(0, 5^2)$ for (C, D), and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.36$ and $\tau = 0.3$.

129 We find that AI usage can have heterogeneous outcomes depending on its type.
130 For example, the use of Omniscient AI can either preserve CI under the feedback
131 incentive structure (Fig. 2A) or promote higher collective accuracy under the niche-
132 expert incentive structure (Fig. 2B). These results are robust across different levels
133 of AI accuracy and can be shown analytically by assuming a sufficiently large group
134 of players (Fig. S2) [SWP: *probably need to swap S2 and S1 so that the current S2*

135 *appears first*].

136 In contrast, the use of Chatbot AI prevents the emergence of CI, thereby resulting
137 in lower levels of collective accuracy (Fig. 2C, D). Under the feedback incentive
138 structure, the use of Chatbot AI leads to a loss of diversity in the population’s
139 interest, making it impossible to account for effect of some causal factors [SWP:
140 Supp]. This causes an increase in collective variance and stochastic decrease in
141 collective accuracy [SWP: Supp]. We note that the loss of interest diversity may
142 arise even in the absence of AI in finite populations [SWP: Supp], but the use of
143 Chatbot AI makes the group more vulnerable to such loss [SWP: Supp]. Under
144 the niche-expert incentive structure, the population becomes overly reliant on AI,
145 causing the collective prediction to inherit the AI’s bias (Fig. S1). This leads to
146 an increase in collective bias and a decrease in collective accuracy [SWP: Supp].
147 Interestingly, we find that the niche-expert structure can prevent the use of AI under
148 specific assumptions about the relationship between the true coefficient and AI bias
149 (Fig. S2). In this case, the population completely excludes AI assistance, allowing
150 the CI to emerge eventually. However, this phenomenon can substantially delay the
151 emergence of CI [SWP: Supp].

152 To investigate whether the group can retain its predictive ability when AI assis-
153 tance suddenly becomes unavailable at any given time, we quantify counterfactual
154 collective accuracy based on unrevised predictions. Across all scenarios, we find that
155 the counterfactual collective accuracy is lower than the collective accuracy of the
156 corresponding population.

157 In summary, an omniscient AI can promote CI, but AI operating with limited
158 information may prevent the emergence CI. Moreover, reliance on AI can limit inher-
159 ent, collective accuracy of the population, raising concerns about future AI usage.
160 These findings underscore the need for incentive structures that can mitigate the
161 potential risks associated with AI assistance.

162 **Directly incentivizing or penalizing AI use can additionally** 163 **harm CI**

164 Given the strong influence of AI on collective outcomes, a straightforward interven-
165 tion is to directly incentivize or penalize the use of AI. To investigate whether such
166 direct incentives or penalties help emergence of CI, we modify the two incentive
167 structures by adding a payoff term proportional to individual reliance on AI:

$$\text{Feedback/Niche-expert} + \lambda\beta_A \quad (1)$$

168 where λ represent the strength of incentive ($\lambda > 0$) or penalty ($\lambda < 0$).

169 Directly incentivizing or penalizing AI use further reduce the collective accuracy
170 (Fig. 3C, E, G), except when Omniscient AI is used under the feedback structure
171 (Fig. 3A). This is because incentivizing or penalizing AI usage pushes the group to
172 always ($\beta = 1$) or never ($\beta = 0$) rely on AI, respectively (Fig. 3B, D, F, H). In some
173 cases, when AI bias is relatively large and the penalty is weak, penalizing AI use can

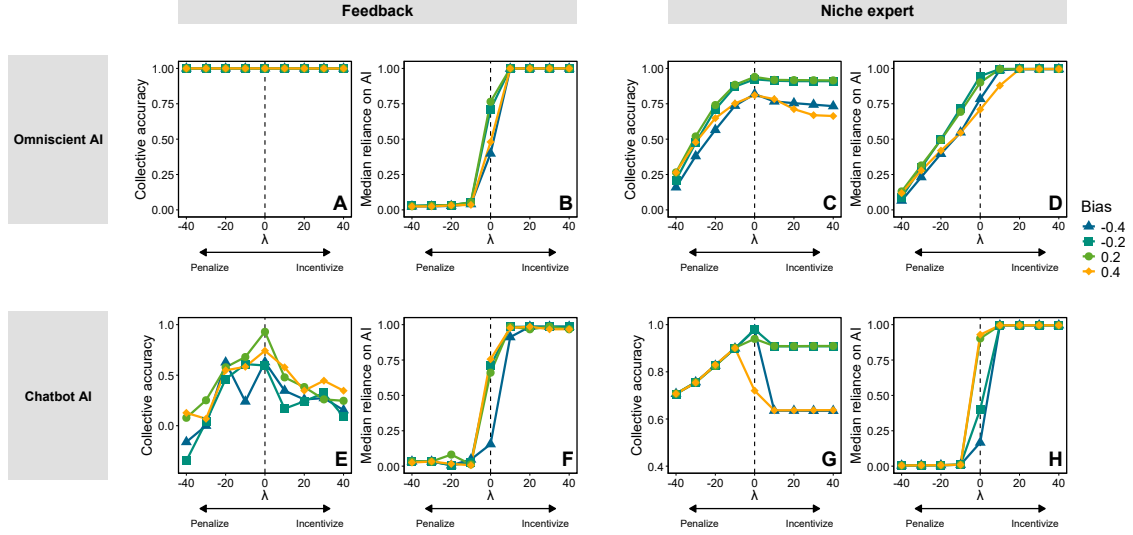


Figure 3: **Collective accuracy under the influence of direct intervention on AI use.** Collective accuracy and median reliance on AI under varying strengths of intervention (λ) applied to the incentive structure. Positive λ represents an incentive for AI use, whereas negative λ represents a penalty for AI use. In each panel, the vertical dashed line indicates the original incentive structure ($\lambda = 0$). (A, B) Omniscient AI under the modified feedback structure, (C, D) Omniscient AI under the modified niche-expert structure, (E, F) Chatbot AI under the modified feedback structure, and (G, H) Chatbot AI under the modified niche-expert structure. Four AI bias levels are shown: $b_i = \pm 0.2$, representing higher-accuracy AI, and $b_i = \pm 0.4$, representing lower-accuracy AI. Collective accuracy and median reliance on AI are averaged over generations 190,000–200,000. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$ for (A–D), $c_A \sim \mathcal{N}(0, 5^2)$ for (E–H), and $\beta_A \sim U(0, 1)$. AI parameter: $\tau = 0.3$.

174 achieve higher collective accuracy than the original incentive structure. However, in
 175 most cases, penalizing AI use imposes strong selection against AI reliance, pushing
 176 the group toward lower stationary collective accuracy. Although the group with
 177 Chatbot AI under feedback structure shows stochastic breakdown of CI which doesn't
 178 result in stationary collective accuracy, incentivizing or penalizing AI use further
 179 strengthens the breakdown (Fig. 3E). This result suggests that simply incentivizing
 180 or penalizing the AI use may exacerbate rather than resolving the problem of AI.

181 **Balanced incentives mitigate AI-induced failures of CI**

182 Neither the original incentive structures nor simple interventions targeting AI use
 183 are sufficient to promote CI. What incentive structure, then, would work in the
 184 presence of AI? Focusing on the Chatbot AI, in which proposed incentive structures

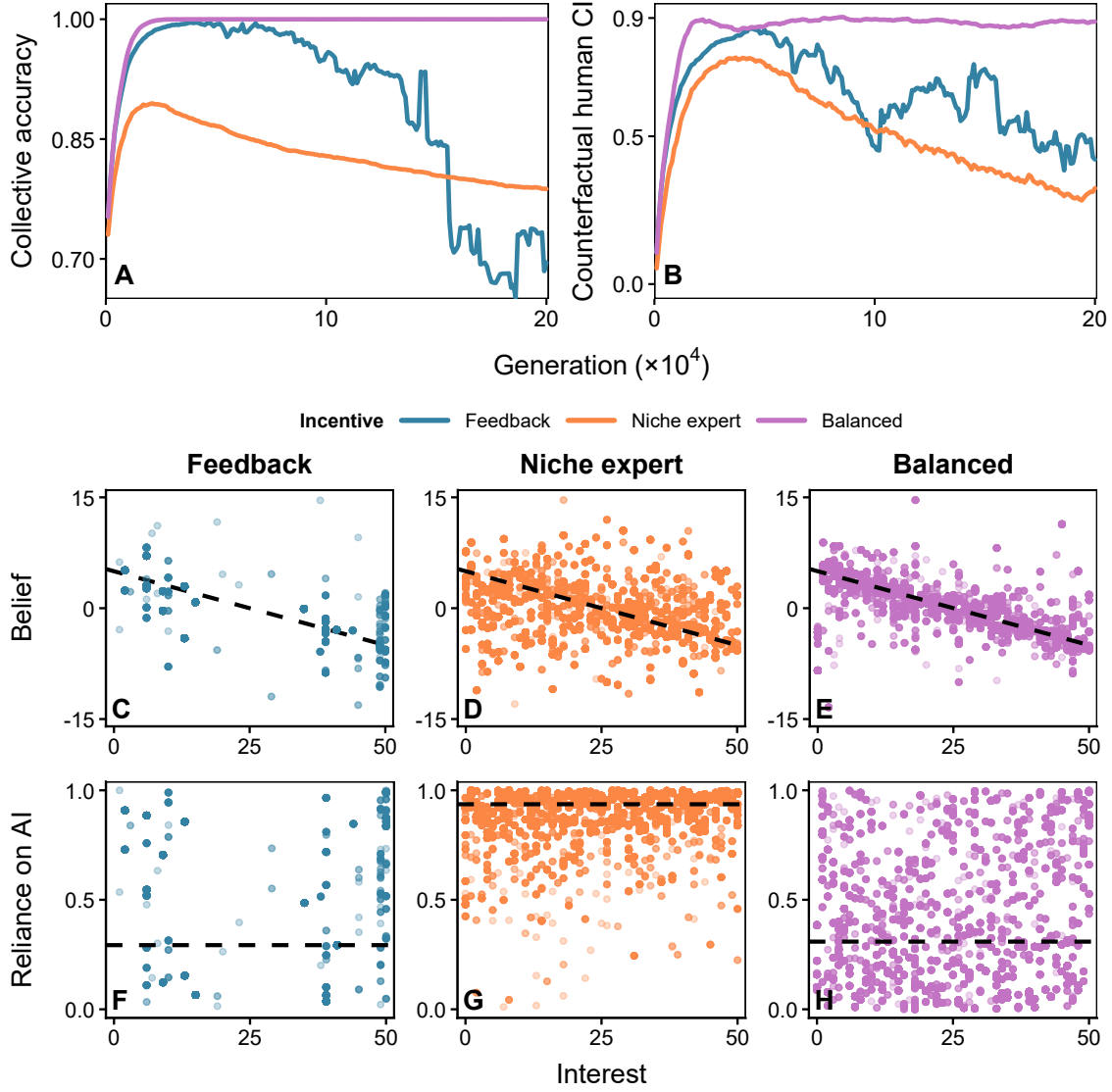


Figure 4: **Balanced structure can promote collective intelligence** Trajectories of collective accuracy (A) and counterfactual human CI (B) under three incentive structures (feedback, niche-expert, and balanced) with Chatbot AI. Distributions of individual interest and belief (C-E) and of interest and reliance on AI (F-H) after 200,000 generations. Each point represents a player. The dashed lines in (C-E) represent the true coefficients (α_i) of the prediction target, and the dashed horizontal lines in (F-H) indicate the median reliance on AI. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i = 5 - 0.2i$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.4$ and $\tau = 0.3$.

fail to promote accurate CI, we investigate whether a better structure can overcome the risks introduced by AI. Here, we propose a “balanced” structure, defined as the

187 simple average of the feedback and niche expert incentives.

188 Core idea behind the balanced structure is to maintain interest diversity while
189 preventing the collective bias of the group. The feedback structure is prone to in-
190 terest diversity loss, whereas can strongly prevent collective bias by incentivizing
191 players whose predictions lie opposite to the collective prediction. By contrast, the
192 niche-expert structure can maintain interest diversity because it directly incentivizes
193 minority interests. However it cannot prevent collective bias because, within the
194 same factor of interest, individuals are rewarded only according to their individual
195 predictive performance, which homogenize the predictions.

196 By integrating both structure, the balanced structure can promote accurate CI by
197 maintaining both interest diversity and preventing collective bias (Fig. 4, S3). Com-
198 pare to the feedback and niche-expert, players are both evenly distributed through
199 factors, and beliefs are close to the real coefficient (Fig. 4). Moreover, counterfac-
200 tual collective accuracy maintained high compare to both feedback and niche-expert.
201 These patterns are driven by the emergent property of the balanced structure, in
202 which group increases its reliance on AI as the AI becomes more accurate, demon-
203 strating appropriately calibrated AI use.

204 Discussion

205 In this paper, we study how AI and incentive structures affect the emergence of CI.
206 Our model suggests that the effect of AI depends on the type of AI used by the
207 group. AI that can incorporate all relevant causal factors may improve group perfor-
208 mance, whereas AI that relies on users to gather information about the causal factors
209 may impair it. By examining different variants of incentive structures, we further
210 show that direct interventions targeting AI use generally harm group performance.
211 Instead, we suggest balanced structure which can promote CI while also generating
212 appropriately calibrated AI use as an emergent property.

213 Although Omniscient AI can produce group performance that is equal to or bet-
214 ter than that achieved without AI, such AI may be difficult to realize in practice
215 because it needs to be able to observe causal factors independently. Achieving this
216 capability would require hardwares, such as sensors, as well as autonomous data ac-
217 quisition capabilities. For this reason, most AI services currently available are better
218 represented by the Chatbot AI in our model, which highlights the potential risks of
219 AI use for CI.

220 In our model, individuals revise their predictions based on the AI’s assistance. It
221 is therefore unsurprising that AI assistance reduces the diversity of individual pre-
222 dictions, a pattern that has also been observed empirically. However, a reduction in
223 prediction diversity does not necessarily lead to the collapse of CI. The principal risk
224 arises when the incentive structure fails to guide the group toward an appropriate
225 set of strategies. The feedback structure leads to the breakdown of CI because it
226 recruits more players to more important factors, which have larger coefficients. This

characteristic creates a risk of interest-diversity loss in finite populations, a situation in which no player remains interested in certain factors because most players shift toward relatively important factors. Without AI, this occurs when initial individual predictions are not sufficiently diverse or when there is no source of novel ideas, such as mutation. Revising individual predictions with AI assistance exacerbates this problem by pulling individuals’ predictions toward the AI’s prediction, thereby reducing prediction diversity even under the same belief distribution. The niche-expert structure hinders CI because it favors extreme levels of reliance on AI. Because individuals within each interest group are affected only by their own performance, incentives based on individual performance reach their maximum at either full reliance on AI or no reliance at all. This extreme reliance results in suboptimal collective accuracy or substantially delays the emergence of CI.

Because AI already generates substantial benefits as well as costs, organizations increasingly seek to improve collective performance by regulating AI use. Such interventions may take direct forms, including AI-use policies or token-based leaderboards, or may operate indirectly through other incentives, such as productivity rewards or the social stigma associated with AI use. However, our model suggests that directly incentivizing or penalizing AI use is unlikely to induce appropriately calibrated reliance on AI or to promote CI. Incentivizing AI use could lead to overreliance, causing groups to propagate AI biases, whereas penalizing AI use could lead to underreliance, preventing groups from using AI even when it performs well in the relevant domain. Our result instead shows that appropriate AI use and accurate CI can be promoted by focusing on the classic properties of groups that have been emphasized for decades—maintaining diversity and preventing collective bias—rather than on AI use itself.

The consistently limited counterfactual collective accuracy observed across AI types and incentive structures suggests that the presence of AI can degrade human group’s own performance, consistent with empirical evidence. This may reduce collective performance when AI assistance becomes unavailable because of resource constraints, system failure, or model collapse, forcing the group to rebuild CI through further adaptation. The higher counterfactual collective accuracy achieved under the balanced structure indicates that an appropriate incentive structure can mitigate the risk of CI breakdown following the loss of AI assistance.

Our model and analysis have several limitations. First, the model considers only social learning among humans and treats the influence of AI as unidirectional. In reality, humans can learn from AI, while AI systems can also be further trained on human-generated data. Such bidirectional adaptation may further reduce group performance by increasing the similarity between human and AI predictions and thereby weakening their complementarity. Empirical studies of human–AI collaboration likewise suggest that stronger correlations between human and AI predictions can diminish the benefits of collaboration. Second, we model AI errors using factor-specific additive biases and random error. Previous studies of AI bias and error suggest that additive biases may arise from skewed training data, or systematic cal-

270 ibration. However, AI errors may also exhibit more complex structures, including
 271 biased estimates of factor effects and correlations across prediction errors. Further
 272 empirical research is therefore needed to characterize the structure of AI bias and
 273 error, together with modeling approaches that can incorporate these more complex
 274 patterns. Finally, to retain the tractability of established CI models, we assume that
 275 the prediction target is a linear combination of causal factors, which does not hold
 276 for many real-world tasks. In particular, many machine learning methods used in
 277 AI systems are capable of capturing nonlinear relationships. Future research should
 278 therefore investigate how nonlinear relationships between causal factors and predic-
 279 tion targets affect the interaction between AI assistance and the emergence of CI.

280 **Methods**

281 **Prediction target**

282 Following previous models of CI, we model the prediction target as a linear combi-
 283 nation of M independent causal factors:

$$Y = \alpha_0 + \alpha_1 X_1 + \cdots + \alpha_M X_M, \quad (2)$$

284 where the fixed coefficient α_i determines the magnitude and direction of the
 285 effect of factor i , and α_0 is a global intercept. Each causal factor X_i is modeled as an
 286 independent Gaussian random variable with mean zero and standard deviation σ_i .

287 **Players**

288 We consider a well-mixed population $\mathcal{N}$ consisting of N players engaged in a novel
 289 prediction task. In the absence of AI assistance, the strategy of player A is char-
 290 acterized by two variables: interest e_A and belief c_A . Player A observes the causal
 291 factor X_{e_A} and predicts the target by multiplying the observed factor value by their
 292 belief:

$$Y_A = c_A X_{e_A}, \quad (3)$$

293 where e_A can take an integer value from 0 to M . When $e_A = 0$, the player
 294 does not observe any causal factor and instead makes a constant prediction based
 295 solely on their belief. We refer to this as a stubborn strategy because the player's
 296 prediction does not respond to the realized values of the causal factors. For notational
 297 convenience, we assume that $X_0 = 1$.

298 When the group is assisted by AI, each player's strategy additionally includes
 299 reliance on AI, denoted by β_A . We assume that each player consults the AI inde-
 300 pendently and revises their prediction by taking a weighted average of their own
 301 prediction and the AI-generated prediction:

$$Y_A^{\text{re}} = \beta_A Y_A^{\text{AI}} + (1 - \beta_A) Y_A, \quad (4)$$

where β_A ranges from 0 to 1. A value of $\beta_A = 1$ indicates complete reliance on AI, whereas $\beta_A = 0$ indicates that the player does not use the AI prediction.

AI models

We assume that the AI model has been trained on similar tasks and is therefore capable of making predictions based on causal factors. However, because the model is imperfect, its outputs are subject to systematic additive bias and stochastic error. The additive bias associated with factor i is denoted by b_i , and the stochastic error is denoted by ϵ . The AI is neither trained nor updated during the simulation, and its biases therefore remain fixed throughout the simulation. Although the bias is indexed by factor, all b_i , including b_0 , are assigned the same value in the main analyses for simplicity. The model formulation nevertheless allows biases to differ among factors.

We assume that the stochastic error follows a Gaussian distribution with mean zero and standard deviation τ . Each time a player consults the AI, an independent error is generated. Errors are therefore independent across players and across repeated consultations by the same player. Consequently, stochastic AI errors can be attenuated when independently generated predictions are aggregated across players. We therefore held τ constant and varied the magnitude of the additive bias to manipulate the level of AI accuracy across analyses.

We consider two extreme types of AI that differ in the amount of information available for prediction. The first is termed the Omniscient AI, which can observe all M causal factors and incorporate them into its prediction regardless of the user's interest. The prediction generated by the Omniscient AI for player A is expressed as follows:

$$Y_A^{\text{AI}} = \alpha_0 + b_0 + \alpha_1 X_1 + b_1 + \cdots + \alpha_M X_M + b_M + \epsilon. \quad (5)$$

The second is termed the Chatbot AI, which can use only the single causal factor X_{e_A} associated with the player's interest. The prediction generated by the Chatbot AI for player A is expressed as follows:

$$Y_A^{\text{AI}} = \alpha_{e_A} X_{e_A} + b_{e_A} + \epsilon. \quad (6)$$

Aggregation rules

The revised individual predictions are aggregated to form a collective prediction. We use two aggregation rules proposed in previous research, depending on the type of AI used by the group.

When the group is assisted by the Omniscient AI, the AI component of each revised prediction contains information about all M factors, even though the individual's own prediction is based on only one factor. Each revised prediction can

therefore be treated as an estimate of the entire target. Accordingly, we calculate the collective prediction by taking the simple average of the revised individual predictions:

$$\hat{Y}^{\text{re}} = \frac{Y_1^{\text{re}} + Y_2^{\text{re}} + \dots + Y_n^{\text{re}}}{N}. \quad (7)$$

When the group is assisted by the Chatbot AI, both the player’s initial prediction and the AI’s prediction are based on the same causal factor. Each revised prediction therefore contains information about only one factor and can be treated as an estimate of that factor’s contribution to the target. We consequently use clustering aggregation.

Clustering aggregation is performed by grouping individuals according to their interests and calculating the mean prediction within each group. The collective prediction is then obtained by summing the mean predictions of all nonempty groups:

$$\hat{Y}^{\text{re}} = \sum_{i:n_i>0} \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} Y_A^{\text{re}}, \quad (8)$$

where $\mathcal{A}_i$ is the set of players interested in factor i , and $n_i = |\mathcal{A}_i|$ is the number of players in that set.

When the group is not assisted by AI, collective predictions are computed from the unrevised individual predictions throughout the simulation. Both simple averaging and clustering can be used in the absence of AI. For fair comparison, we use the same aggregation rule as in the corresponding AI-assisted condition.

Incentive structures

Players receive payoffs determined by an incentive structure based on their strategies. We consider five incentive structures: feedback structure, niche-expert structure, direct intervention on AI use of these two structures, and a balanced structure.

The feedback structure does not directly reward individuals according to their own prediction accuracy. Instead, it rewards predictions that are aligned with the residual error of the collective prediction and therefore have the potential to move the collective prediction toward the true value. Given player A ’s revised prediction Y_A^{re} and the collective prediction $\hat{Y}^{\text{re}}$, the payoff is expressed as

$$\pi_A = Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}). \quad (9)$$

The niche-expert structure assigns higher payoffs to individuals who make accurate predictions while having an underrepresented interest. Specifically, the squared prediction error of player A is weighted by $\rho_{e_A} = n_{e_A}/n$, the proportion of players interested in the same factor:

$$\pi_A = -\rho_{e_A}(Y_A^{\text{re}} - Y)^2. \quad (10)$$

Previous research has shown that these two incentive structures can produce nearly perfect CI through social learning, except when the niche-expert incentive is combined with simple averaging. They are therefore used as baseline structures for examining how AI assistance affects the emergence of CI.

To examine the effects of directly encouraging or discouraging AI use, we modify the feedback and niche-expert structures by adding a term proportional to the player’s reliance on AI:

$$\pi_A = Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) + \lambda\beta_A. \quad (11)$$

$$\pi_A = -\rho_{e_A}(Y_A^{\text{re}} - Y)^2 + \lambda\beta_A. \quad (12)$$

Here, λ is a weighting coefficient representing the strength of the direct incentive or penalty applied to AI use. A positive value of λ rewards AI reliance, whereas a negative value penalizes it. When $\lambda = 0$, no direct incentive or penalty is applied, and the payoff functions reduce to the original feedback and niche-expert structures.

The balanced structure is defined as the arithmetic mean of the feedback and niche-expert payoff components:

$$\pi_A = \frac{Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) - \rho_{e_A}(Y_A^{\text{re}} - Y)^2}{2}. \quad (13)$$

Simulations

All simulations were conducted using an agent-based model with 50 causal factors ($M = 50$) and 10,000 players ($n = 10,000$). The parameters defining the prediction target were sampled at the beginning of the simulations and subsequently held fixed.

The coefficients α_i were sampled from a uniform distribution on $[-5, 5]$, except in the simulation presented in Figure 4, in which α_i was set to $5 - 0.2i$ for visualization. The standard deviations σ_i were sampled from a uniform distribution on $[0, 3]$.

Players’ initial strategies were also sampled from predefined probability distributions. The distribution of individual belief c_A depended on the aggregation rule. Under simple averaging, each factor’s contribution is scaled by the proportion of players interested in that factor. We therefore sampled beliefs from a Gaussian distribution with mean zero and standard deviation 100, following the scaling used in previous work [13]. Under clustering aggregation, beliefs were sampled from a Gaussian distribution with mean zero and standard deviation 5. Because predictions are averaged separately within each interest group and then summed, no additional scaling of c_A is required. The initial interest of each player was sampled uniformly from the integers between 0 and 50, and reliance on AI was sampled from a uniform distribution on $[0, 1]$.

We assumed that prediction events occurred on a substantially faster timescale than social learning. At each generation, we therefore calculated the expected payoff $\mathbb{E}[\pi_A]$ of each player analytically rather than explicitly sampling prediction events.

399 The expectation was taken over the distributions of the causal factors $X_1, \dots, X_M$
 400 and the stochastic AI error ϵ , conditional on the current population strategies and
 401 the fixed target parameters. Closed form expression of expected payoffs were derived
 402 for all incentive structures and are presented in the Supplementary Methods.

403 Each generation consisted of an expected payoff calculation followed by one op-
 404 portunity for pairwise social learning. Two distinct players, A and B , were randomly
 405 sampled, and player A imitated all three strategies of player B with probability

$$\frac{1}{1 + \exp(s(\mathbb{E}[\pi_A] - \mathbb{E}[\pi_B]))}, \quad (14)$$

406 where s represents the strength of social learning and was set to 50 in all sim-
 407 ulations. We introduced a mutation rate μ , such that with probability μ , player A
 408 did not imitate player B but instead redrew all strategy components from the same
 409 distributions used to initialize the population. The mutation rate was set to zero in
 410 all simulations except those presented in Supplementary Figure 4.

411 Unless otherwise specified, each simulation was run for 200,000 generations. When
 412 a result was represented by a single summary value rather than as a temporal tra-
 413 jectory, the metric was averaged over the final 10,000 generations.

414 To evaluate the dynamics of collective prediction, we defined six summary met-
 415 rics: collective accuracy, collective bias, collective variance, counterfactual collective
 416 accuracy, interest diversity, and median reliance on AI. At each population state,
 417 expectations and variances were evaluated analytically over the distributions of the
 418 causal factors and AI errors rather than estimated from sampled prediction events.
 419 Collective bias and collective variance constitute the bias–variance decomposition of
 420 the group level collective prediction error.

421 Collective accuracy was normalized relative to the baseline prediction α_0 . A neg-
 422 ative value indicates that the collective prediction performs worse than the baseline.
 423 Counterfactual collective accuracy was evaluated by setting $\beta_A = 0$ while retaining
 424 the evolved interests and beliefs of the AI-assisted population. In calculating interest
 425 diversity, we used the convention that $\rho_i \ln \rho_i = 0$ when $\rho_i = 0$ (Table 1).

426 Data and code availability

427 All data are stored in a publicly available GitHub repository ([https://github.com/](https://github.com/parklab-snu/collective-intelligence-with-AI)
 428 [parklab-snu/collective-intelligence-with-AI](https://github.com/parklab-snu/collective-intelligence-with-AI)).

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Table 1: Evaluation metrics used to analyze collective prediction dynamics.

Metric	Definition	Interpretation
Collective accuracy	$1 - \frac{\mathbb{E}[(Y - \hat{Y}^{\text{re}})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$	Accuracy of the revised collective prediction relative to the baseline prediction α_0 .
Collective bias	$\left(\mathbb{E}[Y - \hat{Y}^{\text{re}}]\right)^2$	Squared expected deviation of the revised collective prediction from the true target.
Collective variance	$\text{Var}(Y - \hat{Y}^{\text{re}})$	Variability of the revised collective prediction error across prediction events.
Counterfactual collective accuracy	$1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$	Accuracy obtained after removing AI use from the current population while retaining its evolved interests and beliefs.
Interest diversity	$\exp\left(-\sum_{i=0}^M \rho_i \ln \rho_i\right)$	Effective number of interests represented in the population.
Median reliance on AI	$\text{median}_{A \in N}(\beta_A)$	Median reliance on AI in the population.

433 **Contributions**

434 Jaeseung Kim: Conceptualization, Formal Analysis, Investigation, Methodology,
435 Software, Visualization, Writing - Original Draft Preparation, Writing - Review &
436 Editing. Sang Woo Park: Conceptualization, Investigation, Methodology, Supervi-
437 sion, Writing - Original Draft Preparation, Writing - Review & Editing.

438 **Competing Interests**

439 The authors declare no competing interests.

Supplementary Text

1 Omniscient AI

Omniscient AI assumes that AI can incorporate all causal factors regardless of the player's interest.

1.1 Model

$$Y_A^{AI} = \alpha_0 + b_0 + \alpha_1 X_1 + b_1 + \cdots + \alpha_M X_M + b_M + \epsilon. \quad (S1)$$

For simplicity, prediction of Omniscient AI can be written as:

$$Y^{AI} = Y + B + \epsilon \quad (S2)$$

where Y stands for true prediction target ($Y = \alpha_0 + \alpha_1 X_1 + \cdots + \alpha_M X_M$) and B is sum of all fixed biases ($B = b_0 + b_1 + \cdots + b_M$), ϵ follows normal distribution with zero mean, τ standard deviation with no correlation within or between players. Player A's prediction and revised prediction can be written as follows:

$$Y_A = c_A X_{e_A} \quad (S3)$$

$$Y_A^{re} = \beta_A Y^{AI} + (1 - \beta_A) Y_A \quad (S4)$$

where c_A stands for player A's belief on the factor, e_A stands for the interest, and β_A stands for reliance on AI.

1.2 Aggregation

Because Y_A^{AI} is subject to all M factors, we use simple averaging aggregation for collective prediction.

$$\begin{aligned} \hat{Y}^{re} &= \frac{1}{N} \sum_{j=1}^N Y_j^{re} \\ &= \frac{1}{N} \sum_{j=1}^N \{ \beta_j Y^{AI} + (1 - \beta_j) Y_j \} \\ &= \frac{1}{N} \sum_{j=1}^N \{ \beta_j (Y + B + \epsilon) + (1 - \beta_j) c_j X_{e_j} \} \end{aligned}$$

Reparameterization for simple derivation and interpretability.

$$\bar{\beta} := \frac{1}{N} \sum_{j=1}^N \beta_j \quad (S5)$$

457

$$\bar{\eta} := \frac{1}{N} \sum_{j=1}^N \beta_j \epsilon \quad (\text{S6})$$

458

$$\gamma_i := \frac{1}{N} \sum_{A \in \mathcal{A}_i} (1 - \beta_A) c_A \quad (\text{S7})$$

459 $\bar{\beta}$ can be interpreted as mean reliance on AI, γ stands for the coefficient for the factor
 460 i which is solely based on players' belief.

461 By this reparameterization, we can write $\hat{Y}^{re}$ as follows.

$$\hat{Y}^{re} = \bar{\beta}(Y + B) + \sum_{i=0}^M \gamma_i X_i + \bar{\eta} \quad (\text{S8})$$

462 1.3 Expected payoffs

463 Feedback

$$\delta_0 := \alpha_0 - (\bar{\beta}(\alpha_0 + B) + \gamma_0) \quad (\text{S9})$$

464

$$\delta_i := \alpha_i - (\bar{\beta}\alpha_i + \gamma_i), \quad (i \neq 0) \quad (\text{S10})$$

465

$$Y - \hat{Y}^{re} = \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \quad (\text{S11})$$

466 See equation Eq. (S6) for the $\bar{\eta}$.

467 δ_0 is constant term and δ_i is independent of α_k ($k \neq i$), ϵ , and $\bar{\eta}$.

$$\begin{aligned} \mathbb{E}(\pi_A) &= \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re})] \\ &= \mathbb{E} \left[\left\{ \beta_A \alpha_0 + \beta_A \sum_{i=1}^M \alpha_i X_i + \beta_A B + \beta_A \epsilon + (1 - \beta_A) c_A X_{e_A} \right\} \left\{ \delta_0 + \sum_{i=1}^m \delta_i X_i - \bar{\eta} \right\} \right] \\ &= \beta_A \alpha_0 \delta_0 + \beta_A \sum_{i=1}^M \delta_i \alpha_i \sigma_i^2 + \beta_A B \delta_0 - \frac{1}{N} \beta_A^2 \tau^2 + (1 - \beta_A) c_A \delta_{e_A} \sigma_{e_A}^2 \\ &= \beta_A \left\{ (\alpha_0 + B) \delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2 \right\} + (1 - \beta_A) c_A \delta_{e_A} \sigma_{e_A}^2 - \frac{\beta_A^2 \tau^2}{N} \end{aligned}$$

468 **Niche-expert**

469 If $e_A = 0$, player does not observe causal factor and their unrevised prediction is same
 470 with their belief c_A . Such that, the expected payoff is

$$\begin{aligned}
 \mathbb{E}(\pi_A) &= -\mathbb{E}[\rho_0(Y_A^{re} - Y)^2] \\
 &= -\mathbb{E}[\rho_0\{(\beta_A - 1)Y + \beta_A B + \beta_A \epsilon + (1 - \beta_A)c_A\}^2] \\
 &= -\mathbb{E}\left[\rho_0\left\{(\beta_A - 1)(\alpha_0 - c_A) + \beta_A B + (\beta_A - 1)\sum_{i=1}^M \alpha_i X_i + \beta_A \epsilon\right\}^2\right] \\
 &= -\rho_0\left\{(1 - \beta_A)^2(\alpha_0 - c_A)^2 - 2\beta_A(1 - \beta_A)(\alpha_0 - c_A)B + \beta_A^2 B^2\right. \\
 &\quad \left.+ (1 - \beta_A)^2 \sum_{i \neq 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2\right\}
 \end{aligned}$$

471 Else,

$$\begin{aligned}
 \mathbb{E}(\pi_A) &= -\mathbb{E}[\rho_{e_A}(Y_A^{re} - Y)^2] \\
 &= -\mathbb{E}[\rho_{e_A}\{(\beta_A - 1)Y + \beta_A B + \beta_A \epsilon + (1 - \beta_A)c_A X_{e_A}\}^2] \\
 &= -\mathbb{E}\left[\rho_{e_A}\left\{(\beta_A - 1)\alpha_0 + \beta_A B + (\beta_A - 1)(\alpha_{e_A} - c_A)X_{e_A} + (\beta_A - 1)\sum_{i \neq e_A, 0} \alpha_i X_i + \beta_A \epsilon\right\}^2\right] \\
 &= -\rho_{e_A}\left\{(1 - \beta_A)^2 \alpha_0^2 - 2\beta_A(1 - \beta_A)\alpha_0 B + \beta_A^2 B^2 + (1 - \beta_A)^2(\alpha_{e_A} - c_A)^2 \sigma_{e_A}^2\right. \\
 &\quad \left.+ (1 - \beta_A)^2 \sum_{i \neq e_A, 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2\right\}
 \end{aligned}$$

472 **Incentivizing or penalizing AI use**

473 Because additive term $\lambda\beta$ is not a random variable, we can simply derive the expected
 474 payoff as

$$\mathbb{E}(\pi_{A, Feedback}) = \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re}) + \lambda\beta_A] = \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re})] + \lambda\beta_A. \quad (S12)$$

$$\mathbb{E}(\pi_{A, Niche-expert}) = -\mathbb{E}[\rho_{e_A}(Y_A^{re} - Y)^2 + \lambda\beta_A] = -\mathbb{E}[\rho_{e_A}(Y_A^{re} - Y)^2] + \lambda\beta_A. \quad (S13)$$

475 **1.4 Expected accuracy**

$$\mathbb{E}[(Y - \alpha_0)^2] = \mathbb{E}\left[\left(\alpha_0 + \sum_{i=1}^M \alpha_i X_i - \alpha_0\right)^2\right] = \sum_{i=1}^M \alpha_i^2 \sigma_i^2 \quad (S14)$$

476

$$\mathbb{E}[(Y - \hat{Y}^{re})^2] = \mathbb{E}\left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i - \hat{\eta}\right)^2\right] = \delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \frac{\tau^2}{N^2} \sum_{j=1}^N \beta_j^2 \quad (\text{S15})$$

477

$$\begin{aligned} R &= 1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]} \\ &= 1 - \frac{\delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \frac{\tau^2}{N^2} \sum_{j=1}^N \beta_j^2}{\sum_{i=1}^M \alpha_i^2 \sigma_i^2} \end{aligned}$$

478 2 Chatbot AI

479 2.1 Model

$$Y_A^{AI} = \alpha_{e_A} X_{e_A} + b_{e_A} + \epsilon \quad (\text{S16})$$

480 b_{e_A} stands for factor specific additive bias, ϵ follows normal distribution with zero
 481 mean, τ standard deviation with no correlation within or between players. Player
 482 A's prediction and revised prediction can be written as follows:

$$Y_A = c_A X_{e_A} \quad (\text{S17})$$

483

$$Y_A^{re} = \beta_A Y_A^{AI} + (1 - \beta_A) Y_A \quad (\text{S18})$$

484 where c_A stands for player A's belief on the factor, e_A stands for the interest, and
 485 β_A stands for reliance on AI.

486 2.2 Aggregation

487 For Chatbot AI, clustering aggregation is used for collective prediction.

$$\begin{aligned} \hat{Y}^{re} &= \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} Y_A^{re} \\ &= \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \{\beta_A Y_A^{AI} + (1 - \beta_A) Y_A\} \\ &= \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} [\{\beta_A \alpha_{e_A} + (1 - \beta_A) c_A\} X_{e_A} + \beta_A b_{e_A} + \beta_A \epsilon] \end{aligned}$$

488 Reparameterize for the simplicity.

$$\tilde{c}_i := \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \{\beta_A \alpha_{e_A} + (1 - \beta_A) c_A\} \quad (\text{S19})$$

489

$$\tilde{B}_i := \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \beta_A b_{e_A} \quad (\text{S20})$$

490

$$\bar{B} := \sum_{i=0}^M \tilde{B}_i \quad (\text{S21})$$

491

$$\tilde{\eta}_i := \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \beta_A \epsilon \quad (\text{S22})$$

492

$$\bar{\eta} := \sum_{i=0}^M \tilde{\eta}_i \quad (\text{S23})$$

493 2.3 Expected payoffs

494 Feedback

495

$$\delta_0 := \alpha_0 - \tilde{c}_0 - \bar{B} \quad (\text{S24})$$

496

$$\delta_i := \alpha_i - \tilde{c}_i, \quad (i \neq 0) \quad (\text{S25})$$

$$Y - \hat{Y}^{re} = \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \quad (\text{S26})$$

497 δ_0 is constant term and δ_i is independent of α_k ($k \neq i$), ϵ , and $\bar{\eta}$.

$$\begin{aligned} \mathbb{E}(\pi_A) &= \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re})] \\ &= \mathbb{E} \left[\{(\beta_A \alpha_{e_A} + (1 - \beta_A) c_A) X_{e_A} + \beta_A b_{e_A} + \beta_A \epsilon\} \left\{ \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \right\} \right] \\ &= \{\beta_A \alpha_{e_A} + (1 - \beta_A) c_A\} \delta_{e_A} \sigma_{e_A}^2 + \beta_A b_{e_A} \delta_0 - \frac{1}{n_{e_A}} \beta_A^2 \tau^2 \end{aligned}$$

498 Niche-expert

499 If $e_A = 0$,

$$\begin{aligned} \mathbb{E}(\pi_A) &= -\mathbb{E}[\rho_0(Y_A^{re} - Y)^2] \\ &= -\mathbb{E} \left[\rho_0 \left[\{\beta_A \alpha_0 + (1 - \beta_A) c_A\} + \beta_A b_0 + \beta_A \epsilon - \alpha_0 - \sum_{i=1}^M \alpha_i X_i \right]^2 \right] \\ &= -\rho_0 \left\{ (1 - \beta_A)^2 (c_A - \alpha_0)^2 + 2(1 - \beta_A)(c_A - \alpha_0) \beta_A b_0 + \beta_A^2 b_0^2 \right. \\ &\quad \left. + \sum_{i \neq 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2 \right\} \end{aligned}$$

500 Else,

$$\begin{aligned}
\mathbb{E}(\pi_A) &= -\mathbb{E}[\rho_{e_A}(Y_A^{\text{re}} - Y)^2] \\
&= -\mathbb{E}\left[\rho_{e_A}\left[\{\beta_A\alpha_{e_A} + (1 - \beta_A)c_A\}X_{e_A} + \beta_A b_{e_A} + \beta_A\epsilon - \alpha_0 - \sum_{i=1}^M \alpha_i X_i\right]^2\right] \\
&= -\rho_{e_A}\left\{(1 - \beta_A)^2(c_A - \alpha_{e_A})^2\sigma_{e_A}^2 + \sum_{i \neq e_A, 0} \alpha_i^2 \sigma_i^2\right. \\
&\quad \left.+ \alpha_0^2 + \beta_A^2 b_{e_A}^2 - 2\alpha_0\beta_A b_{e_A} + \beta_A^2 \tau^2\right\}
\end{aligned}$$

501 Incentivizing or penalizing AI use

502 Because additive term $\lambda\beta$ is not a random variable, we can simply derive the expected
503 payoff as

$$\mathbb{E}(\pi_{A, \text{Feedback}}) = \mathbb{E}\left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) + \lambda\beta_A\right] = \mathbb{E}\left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}})\right] + \lambda\beta_A. \quad (\text{S27})$$

$$\mathbb{E}(\pi_{A, \text{Niche-expert}}) = -\mathbb{E}\left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2 + \lambda\beta_A\right] = -\mathbb{E}\left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2\right] + \lambda\beta_A. \quad (\text{S28})$$

504 Balanced

505 Expected payoff of the balanced structure can also be driven by arithmetic mean of
506 expected payoff of feedback and niche-expert.

$$\mathbb{E}(\pi_A) = \mathbb{E}\left[\frac{Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) - \rho_{e_A}(Y_A^{\text{re}} - Y)^2}{2}\right] = \frac{\mathbb{E}\left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}})\right] - \mathbb{E}\left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2\right]}{2} \quad (\text{S29})$$

507 2.4 Expected accuracy

$$\mathbb{E}[(Y - \alpha_0)^2] = \mathbb{E}\left[\left(\alpha_0 + \sum_{i=1}^M \alpha_i X_i - \alpha_0\right)^2\right] = \sum_{i=1}^M \alpha_i^2 \sigma_i^2 \quad (\text{S30})$$

$$\mathbb{E}[(Y - \hat{Y})^2] = \mathbb{E}\left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i - \hat{\eta}\right)^2\right] = \delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \sum_{i=0}^M \frac{\tau^2}{n_i^2} \sum_{A \in \mathcal{A}_i} \beta_A^2 \quad (\text{S31})$$

$$\begin{aligned}
R &= 1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]} \\
&= 1 - \frac{\delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \sum_{i=0}^M \frac{\tau^2}{n_i^2} \sum_{A \in \mathcal{A}_i} \beta_A^2}{\sum_{i=1}^M \alpha_i^2 \sigma_i^2}
\end{aligned}$$

510 3 Evolutionary dynamics

511 For sufficiently large population ($N \rightarrow \infty$), we can approximate the deterministic
512 evolutionary dynamics using the replicator equation. Let λ denote a strategy, p_λ its
513 frequency, and $u_\lambda = \mathbb{E}(\pi_\lambda)$ its expected payoff. The replicator equation is

$$\frac{dp_\lambda}{dt} = p_\lambda(u_\lambda - \bar{u}), \quad (\text{S32})$$

514 where $\bar{u} = \sum_\lambda p_\lambda u_\lambda$. Therefore, at an internal stationary state, all strategies
515 with positive frequency have the same expected payoff. A boundary state can also
516 be stationary because a strategy with exactly zero frequency cannot increase under
517 the replicator equation. Thus, stability of a boundary state needs to be considered
518 separately by examining invasion by a rare strategy.

519 3.1 Omniscient AI

520 3.1.1 Feedback

521 When the population size approaches infinity, the effect of stochastic error on the
522 collective prediction vanishes ($\bar{\eta} \rightarrow 0$) and the effect of stochastic error in the ex-
523 pected payoff, $-\beta^2 \tau^2 / N$, also vanishes. Following prior research [13], we define the
524 Lyapunov function as the expected error of collective prediction

$$V = \mathbb{E} \left[(Y - \hat{Y}^{re})^2 \right]. \quad (\text{S33})$$

525 The function V is non-negative and equals zero when the collective prediction is
526 identical to the target. To examine how V changes under the evolutionary dynamics,
527 we take its time derivative:

$$\begin{aligned}
\frac{dV}{dt} &= -2\mathbb{E} \left[(Y - \hat{Y}^{re}) \frac{d\hat{Y}^{re}}{dt} \right] \\
&= -2\mathbb{E} \left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i \right) \left\{ \frac{d\bar{\beta}}{dt} (Y + B) + \sum_{i=0}^M \frac{d\gamma_i}{dt} X_i \right\} \right] \\
&= -2 \left[\left\{ (\alpha_0 + B) \delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2 \right\} \frac{d\bar{\beta}}{dt} + \delta_0 \frac{d\gamma_0}{dt} + \sum_{i=1}^M \delta_i \sigma_i^2 \frac{d\gamma_i}{dt} \right].
\end{aligned}$$

528 For notational convenience, we define

$$K = (\alpha_0 + B) \delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2. \quad (\text{S34})$$

529 Although K changes with the population state, it is common to all strategies at a
530 given state and can therefore be treated as a constant with respect to the strategy
531 index λ . We also define

$$G_e = \begin{cases} \delta_0, & e = 0, \\ \delta_e \sigma_e^2, & e \neq 0. \end{cases} \quad (\text{S35})$$

532 By the definitions of $\bar{\beta}$ and γ_i and the replicator equation,

$$\frac{d\bar{\beta}}{dt} = \sum_{\lambda} p_{\lambda} (u_{\lambda} - \bar{u}) \beta_{\lambda} = \text{Cov}(\beta, u). \quad (\text{S36})$$

533 Similarly, the terms involving γ_i can be combined as

$$\delta_0 \frac{d\gamma_0}{dt} + \sum_{i=1}^M \delta_i \sigma_i^2 \frac{d\gamma_i}{dt} = \text{Cov}((1 - \beta) c G_e, u). \quad (\text{S37})$$

534 Therefore,

$$\begin{aligned}
\frac{dV}{dt} &= -2 \{ K \text{Cov}(\beta, u) + \text{Cov}((1 - \beta) c G_e, u) \} \\
&= -2 \text{Cov}(K\beta + (1 - \beta) c G_e, u).
\end{aligned}$$

535 For the feedback structure in the limit $N \rightarrow \infty$, the expected payoff of strategy
536 λ is

$$u_{\lambda} = K \beta_{\lambda} + (1 - \beta_{\lambda}) c_{\lambda} G_{e_{\lambda}}. \quad (\text{S38})$$

537 Which is same with Thus,

$$\frac{dV}{dt} = -2 \text{Cov}(u, u) = -2 \text{Var}(u) \leq 0. \quad (\text{S39})$$

Therefore, the expected error of collective prediction cannot increase during the evolutionary process and decreases whenever strategies differ in their expected payoffs. The derivative becomes zero when the variance of expected payoffs is zero, $\text{Var}(u) = 0$. This condition corresponds to equal expected payoffs among the strategies present in the population. A boundary state in which the entire population consists of a single strategy can also satisfy $\text{Var}(u) = 0$ even when $V > 0$. However, such a state is not evolutionarily stable under the feedback structure. If the collective prediction differs from the target Y , a nonzero residual remains. By the definition of the feedback payoff, a novel strategy whose prediction aligns with the direction of this residual obtains a higher expected payoff than the resident strategy and can therefore invade the population. Thus, an imperfect boundary equilibrium with $V > 0$ is unstable to the introduction of novel predictions.

Consequently, the evolutionary dynamics continue to reduce the collective prediction error until the population reaches the stable state with zero residual, $V = 0$. At this state, the collective prediction equals the target, $\hat{Y}^{re} = Y$, and the collective accuracy reaches $R = 1$.

3.1.2 Niche-expert

Unlike the stochastic error in the collective prediction, stochastic error in the niche-expert payoff is evaluated at the individual level and does not vanish as population size increases. Therefore, we retain $\tau > 0$ in the following derivation.

We first consider strategies having the same interest by optimizing the expected payoff with respect to c and β . For a nonzero interest e , the expected payoff is strictly concave with respect to c for $\beta < 1$ and $\sigma_e > 0$, and is maximized at

$$c_e^* = \alpha_e. \quad (\text{S40})$$

After substituting $c_e^* = \alpha_e$, the payoff is strictly concave with respect to β . Its unconstrained stationary point is

$$\tilde{\beta}_e = \frac{\alpha_0^2 + \alpha_0 B + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2}{(\alpha_0 + B)^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S41})$$

Because β is constrained between 0 and 1, the optimal β_e^* is $\tilde{\beta}_e$ when $0 < \tilde{\beta}_e < 1$, and otherwise lies at the corresponding boundary, $\beta_e^* = 0$ or $\beta_e^* = 1$. Thus, the boundary cases do not require a separate evolutionary argument; they only change the optimal prediction and its minimum expected squared error. When $\beta_e^* = 1$, the revised prediction is independent of c_e , so different values of c_e are selectively neutral and can be treated as a single payoff-equivalent strategy class.

Let L_e denote the minimum expected individual error, $\mathbb{E}[(Y_A^{re} - Y)^2]$, within interest e . For an internal optimum,

$$L_e = \frac{B^2 \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2 \left(\alpha_0^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 \right)}{(\alpha_0 + B)^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S42})$$

571 If the optimum is located at $\beta_e^* = 0$ or $\beta_e^* = 1$, L_e is instead obtained by evaluating
 572 the expected squared prediction error at the corresponding boundary.

573 For interest 0, optimization with respect to c gives

$$c_0^*(\beta) = \alpha_0 - \frac{\beta}{1 - \beta} B. \quad (\text{S43})$$

574 Substituting this into the expected payoff gives a strictly concave function of β ,
 575 with the unique optimum

$$\beta_0^* = \frac{\sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S44})$$

576 Therefore,

$$c_0^* = \alpha_0 - \frac{B}{\tau^2} \sum_{j=1}^M \alpha_j^2 \sigma_j^2, \quad (\text{S45})$$

577 and

$$L_0 = \frac{\tau^2 \sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S46})$$

578 We denote the optimal strategy, or the payoff-equivalent strategy class when
 579 $\beta_e^* = 1$, by S_e . A state containing a suboptimal strategy within an interest can
 580 be stationary if the optimal strategy is absent, but it is unstable against invasion
 581 because the optimal strategy has a strictly higher payoff at the same ρ_e . Therefore,
 582 a stable state can contain only S_e within each represented interest.

583 We next consider the frequencies of the $M + 1$ optimal strategies. Let ρ_e denote
 584 the frequency of S_e . Its expected payoff is

$$u_e = -\rho_e L_e. \quad (\text{S47})$$

585 Assuming $L_e > 0$ for all interests, the equal-payoff condition of the replicator
 586 equation at an internal stationary state gives

$$\rho_0^* L_0 = \rho_1^* L_1 = \cdots = \rho_M^* L_M. \quad (\text{S48})$$

587 Together with $\sum_{e=0}^M \rho_e^* = 1$, this uniquely determines

$$\rho_e^* = \frac{D}{L_e}, \quad D = \left(\sum_{j=0}^M \frac{1}{L_j} \right)^{-1}. \quad (\text{S49})$$

588 This internal stationary state is locally stable. At equilibrium, all represented
 589 interests have the same expected payoff, with $\rho_e^* L_e = D$. If the frequency of one
 590 interest increases slightly above its equilibrium value, its payoff decreases because
 591 the niche-expert payoff is proportional to $-\rho_e L_e$. Conversely, if the frequency of
 592 an interest decreases below its equilibrium value, its payoff increases. The replicator
 593 dynamics therefore generate a restoring force that reduces small deviations in interest
 594 frequencies and drives the population back toward the internal stationary state.

595 If one or more interests are absent, the resulting boundary state can be stationary
 596 under the replicator equation, but it is unstable against invasion. The payoff of a rare
 597 absent interest approaches zero as $\rho_e \rightarrow 0$, whereas the mean payoff of the resident
 598 population is negative when $L_e > 0$, so the rare interest increases when introduced
 599 at sufficiently small frequency.

600 Thus, there exists a unique stable full-support equilibrium in terms of the optimal
 601 prediction strategies and their frequencies. Whether the constrained optimum has
 602 $\beta_e^* = 0$, $0 < \beta_e^* < 1$, or $\beta_e^* = 1$ does not change the frequency argument, because
 603 these cases differ only through the resulting value of L_e . When $\beta_e^* = 1$, the strategy
 604 is selectively neutral with respect to c_e , but its prediction, payoff, and equilibrium
 605 interest frequency remain uniquely determined.

606 This equilibrium also allows a direct comparison with the expected accuracy of
 607 the AI. Using Y^{AI} itself without aggregation as the prediction gives

$$\mathbb{E} [(Y - Y^{\text{AI}})^2] = B^2 + \tau^2. \quad (\text{S50})$$

608 For every interest, complete reliance on AI, $\beta = 1$, is a feasible strategy. There-
 609 fore, because L_e is the minimum individual expected squared prediction error within
 610 interest e ,

$$L_e \leq B^2 + \tau^2. \quad (\text{S51})$$

611 Moreover, for interest 0,

$$L_0 = \frac{\tau^2 \sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2} < \tau^2 \leq B^2 + \tau^2. \quad (\text{S52})$$

612 Because the stable equilibrium has $\rho_0^* > 0$,

$$\sum_{e=0}^M \rho_e^* L_e < B^2 + \tau^2. \quad (\text{S53})$$

613 The collective prediction is the average of individual revised predictions. By
 614 convexity of the squared error,

$$\mathbb{E} [(Y - \hat{Y}^{re})^2] \leq \sum_{e=0}^M \rho_e^* L_e. \quad (\text{S54})$$

615 Therefore,

$$\mathbb{E}[(Y - \hat{Y}^{re})^2] < \mathbb{E}[(Y - Y^{AI})^2]. \quad (\text{S55})$$

616 Because the collective accuracy and AI accuracy use the same normalization by
 617 $\mathbb{E}[(Y - \alpha_0)^2]$, the stationary collective accuracy under the niche-expert structure is
 618 strictly greater than the expected accuracy of the AI prediction.

619 Although the niche-expert structure improves collective prediction relative to the
 620 AI, the stationary collective accuracy does not reach one when $\tau > 0$. For a nonzero
 621 factor i , the residual coefficient of the stationary collective prediction can be written
 622 as

$$\delta_i^* = \alpha_i \{1 - \bar{\beta}^* - \rho_i^*(1 - \beta_i^*)\} = \alpha_i \sum_{e \neq i} \rho_e^*(1 - \beta_e^*). \quad (\text{S56})$$

623 At the stable equilibrium, $\rho_0^* > 0$. In addition, $\tau > 0$ implies $\beta_0^* < 1$. Therefore,
 624 for every factor with $\alpha_i \neq 0$,

$$\delta_i^* \neq 0. \quad (\text{S57})$$

625 Because $\sum_{i=1}^M \alpha_i^2 \sigma_i^2 > 0$, at least one factor contributes positive prediction error.
 626 Hence,

$$\mathbb{E}[(Y - \hat{Y}^{re})^2] > 0, \quad (\text{S58})$$

627 and consequently the stationary collective accuracy satisfies $R^* < 1$.

628 **3.2 Chatbot AI**

629 **3.2.1 Feedback**

630 With Chatbot AI, the effect of stochastic error on the collective prediction depends
 631 on n_i , requiring the additional assumption that $n_i \rightarrow \infty$ for every interest for the
 632 stochastic error to vanish. However, Under this assumption, the loss of interest
 633 diversity cannot occur, and therefore the stochastic breakdown of collective accuracy
 634 observed in finite populations cannot be analytically captured. Instead, by assuming
 635 that each interest is represented by infinitely many players, we can show that the
 636 expected collective prediction error monotonically decreases until perfect accuracy
 637 is achieved. Because finite population dynamics are more relevant to the empirical
 638 populations considered in this study, we retain finite population simulations for the
 639 main result.

640 **3.2.2 Niche-expert**

641 We first consider strategies having the same interest by optimizing the expected
 642 payoff with respect to c and β . For a nonzero interest e , the expected payoff is
 643 strictly concave with respect to c for $\beta < 1$ and $\sigma_e > 0$, and is maximized at

$$c_e^* = \alpha_e. \quad (\text{S59})$$

644 After substituting $c_e^* = \alpha_e$, the payoff is strictly concave with respect to β . Its
 645 unconstrained stationary point is

$$\tilde{\beta}_e = \frac{2\alpha_0 b_e}{2b_e^2 + 2\tau^2}. \quad (\text{S60})$$

646 Because β is constrained between 0 and 1, the optimal β_e^* is $\tilde{\beta}_e$ when $0 < \tilde{\beta}_e < 1$,
 647 and otherwise lies at the corresponding boundary, $\beta_e^* = 0$ or $\beta_e^* = 1$. Because we
 648 assumes AI model with moderate accuracy, b_e and τ is relatively smaller than α_0
 649 which makes $\tilde{\beta}_e > 1$ when $\alpha_0 b_e > 0$ and $\tilde{\beta}_e < 0$ when $\alpha_0 b_e < 0$. These boundary
 650 cases lead to extreme reliance on AI or complete exclusion of AI which both hinders
 651 the emergence of CI. Only when the prediction task is centered near zero and AI is
 652 relatively inaccurate ($\alpha_0 b_e < b_e^2 + \tau^2$), and $\alpha_0 b_e > 0$, the internal optimum β_e^* can be
 653 exist.

654 For interest 0, optimization with respect to c gives

$$c_0^*(\beta) = \alpha_0 - \frac{\beta}{1 - \beta} b_0. \quad (\text{S61})$$

655 Substituting this into the expected payoff gives a strictly concave function of β , with
 656 the unique optimum

$$\beta_0^* = 0. \quad (\text{S62})$$

Supplementary Figure

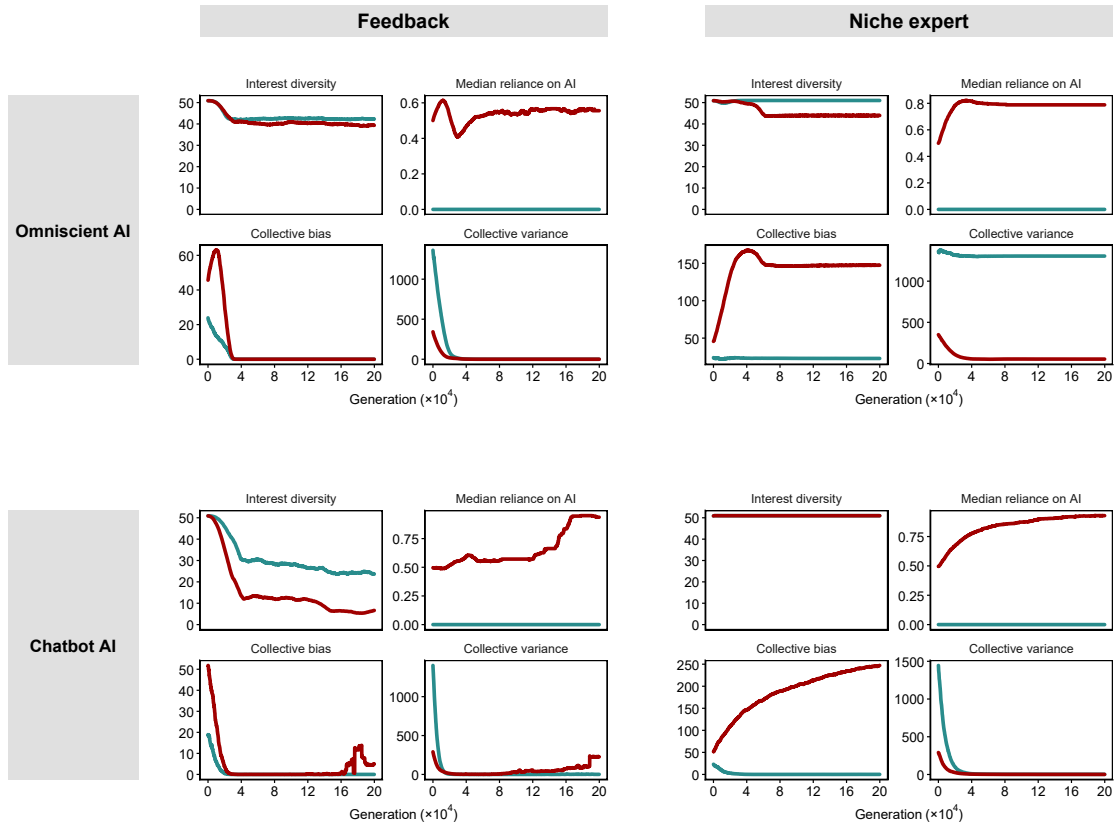


Figure S1: Caption. Text.

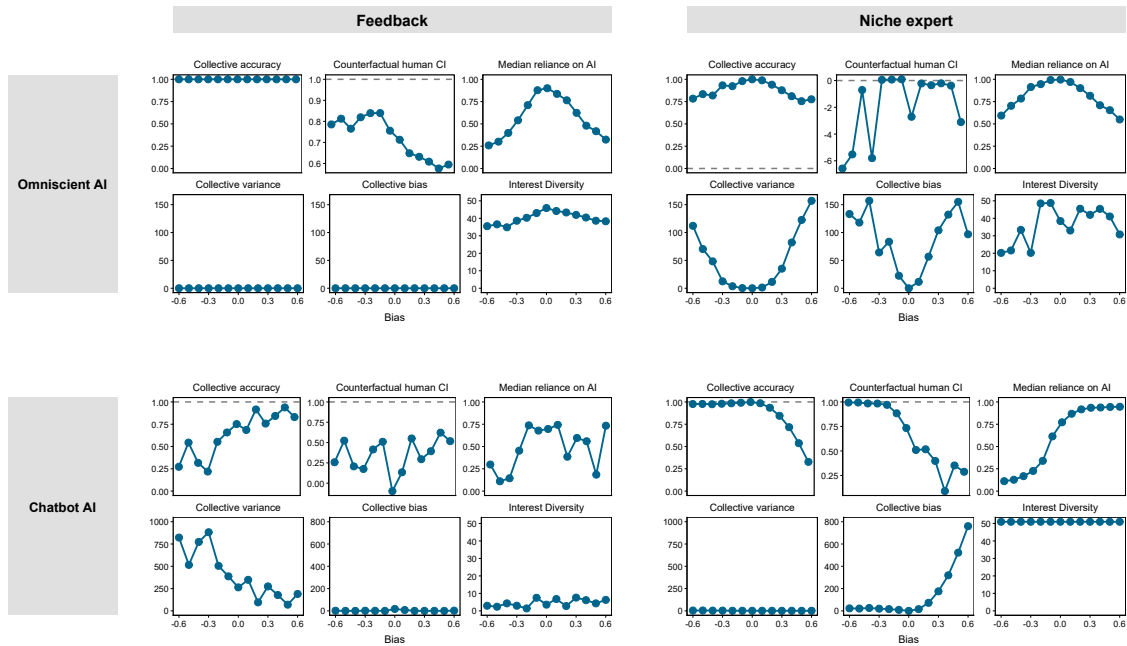


Figure S2: Caption. Text.

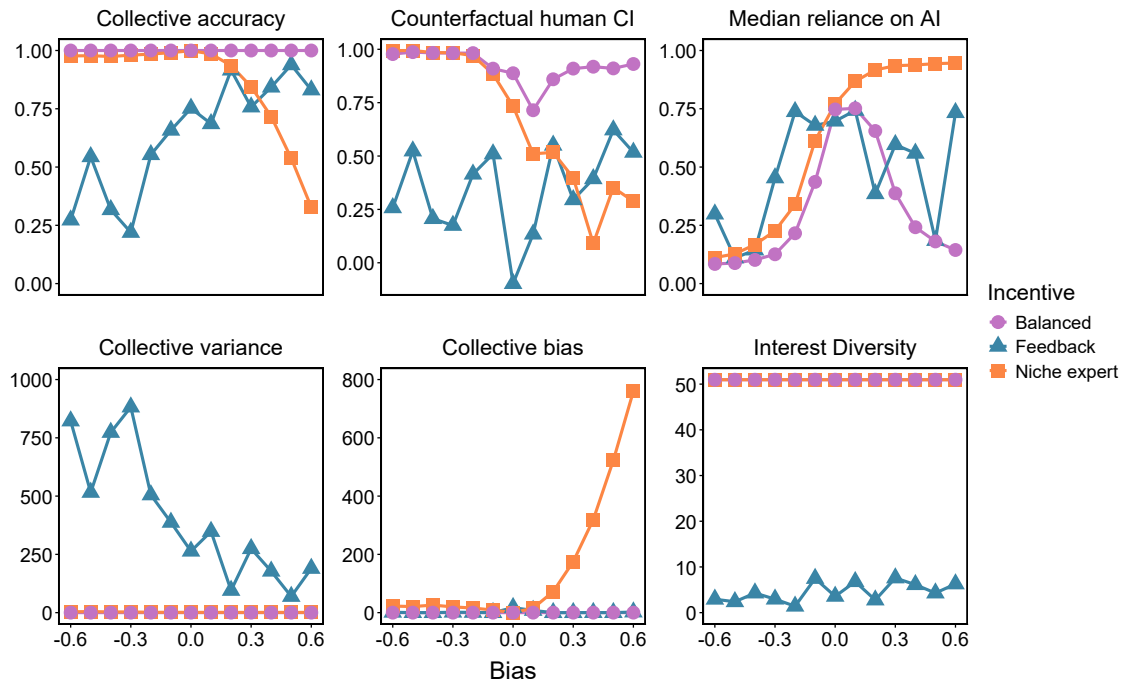


Figure S3: **Caption.** Text.

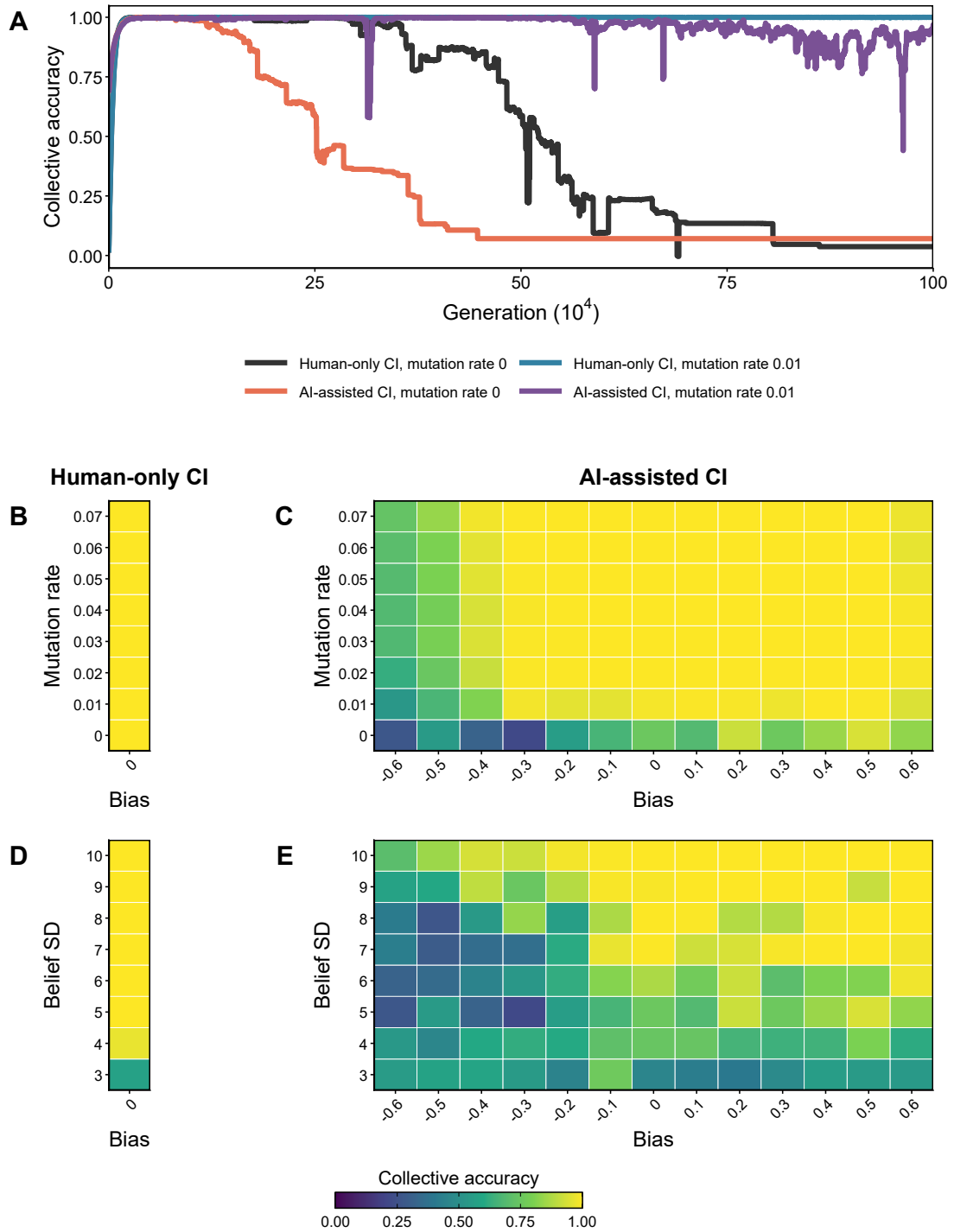


Figure S4: **Caption.** Text.

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